BEMS GROUP CONNECT · OFFICIAL TECHNICAL SPECIFICATION

AI Automation & n8n Engineer Integration Guide (v2.0.0)

Architecture: 100% Staged-Action Governance (Zero Direct AI Writes · Mandatory Human Button Confirmation)
Coverage: All 16 BEMI AI Tool Endpoints · 45+ Sub-Actions · Authentication · Payloads · Staged Envelopes

BEMS ERP — AI Automation & n8n Engineer Integration Guide

Version: 2.0.0 (Universal Human-in-the-Loop Staged-Action Architecture)

Target Environments: Production (`https://bemserp.work`) | Local Development (`http://localhost:3000`)

Authentication Standard: Bearer Token / `x-api-key` Service Header + Optional Actor Attribution (`x-actor-id` / `x-actor-email`)

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1. Executive Architecture & Mandatory Staged-Action Governance

1.1 Zero Direct AI Writes (100% Staged Execution)

In BEMS Group Connect, **the AI agent and n8n automation workflows never mutate live database records directly**.

- `GET` **requests** across `/api/ai/tools/*` are read-only and immediately return live ERP data for reasoning, financial analysis, and context gathering.
- `POST` **requests** across `/api/ai/tools/*` **exclusively stage a proposal** in the `StagedAction` table (`status: "AWAITING_CONFIRMATION"` , `staged: true`) and return an interactive preview payload (`actionId` , `summary` , `preview` , `confirmUrl` , `cancelUrl` , `expiresAt`).
- **Database execution occurs ONLY when an authenticated human user clicks the confirmation button** in the BEMS Web Dashboard / BEMI Copilot Drawer (`POST /api/ai/staged-actions/{actionId}/confirm`).

1.2 Prohibited Bypass Vectors

To satisfy strict audit controls:

1. **No Auto-Confirm Flags:** Flags such as `auto_confirm: true` or `confirmed_by_gmd: true` in tool request payloads are ignored. Every tool `POST` returns a staged proposal.

2. **No Service-Token Confirmation:** The confirmation endpoint (`POST /api/ai/staged-actions/{id}/confirm`) and cancellation endpoint (`POST /api/ai/staged-actions/{id}/cancel`) **reject** requests carrying `x-api-key` or `Authorization: Bearer <BEMS_AI_SERVICE_TOKEN>` with `HTTP 403 Forbidden`. Only a browser/mobile human session cookie can confirm or cancel a staged action.
3. **No Conversational Text Confirmation:** Typing *"yes"*, *"approve"*, or *"confirm"* in the chat does not execute a staged action. The human must click the **Confirm & Execute** button on the staged action card.
4. **Strict 3-Role Architecture:** Only three roles exist in BEMS: `SUPER_ADMIN` (stored as `SYSTEM_ADMIN` in DB), `GMD` (The Group Managing Director), and `STAFF`. Legacy roles (`FINANCE`, `COO`, `HR`, `DEPT_HEAD`, `VERIFIER`, `DIRECTOR`) do not exist.

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2. Connection & Authentication Headers

Every n8n **HTTP Request** node calling `/api/ai/tools/*` must include your `BEMS_AI_SERVICE_TOKEN` via `Authorization: Bearer <token>` or `x-api-key: <token>`. You may also pass `x-actor-email` or `x-actor-id` so the staged proposal is attributed to the specific executive interacting with the bot.

Required HTTP Headers

```http
Authorization: Bearer <BEMS_AI_SERVICE_TOKEN>
Content-Type: application/json
x-actor-email: gmd@bemsgroup.com
```

(Alternative auth header: `x-api-key: <BEMS_AI_SERVICE_TOKEN>`)

Standard Staged-Action Response Envelope (Returned by ALL `POST /api/ai/tools/*` Calls)

```JSON
{
  "ok": true,
  "staged": true,
  "status": "AWAITING_CONFIRMATION",
  "message": "Action has been staged. An authorized user must click Confirm in the
dashboard.",
  "actionId": "sa_9df4c2b76ab34bbd",
  "actionType": "approval",
  "summary": "Approve REQ-2026-0005 (Starlink High-Speed Router Replacement) for ₦150,000",
  "preview": {
    "operation": "Approve Fund Request",
    "requestNumber": "REQ-2026-0005",
    "amount": "₦150,000"
  },
  "payloadHash": "sha256:8f9d2a...",
  "expiresAt": "2026-11-06T11:15:00.000Z",
  "confirmUrl": "/api/ai/staged-actions/sa_9df4c2b76ab34bbd/confirm",
  "cancelUrl": "/api/ai/staged-actions/sa_9df4c2b76ab34bbd/cancel"
}
```

3. Master Discovery & Context Endpoints

3.1 Master Tool Index

- **Method:** `GET`
- **Endpoint:** `/api/ai/tools`
- **Purpose:** Returns the live registry of all 16 BEMI AI tools, their endpoints, supported sub-actions, and staging rules (`stages: true | directExecute: false`).

3.2 Unified Conglomerate Context Snapshot

- **Method:** `GET`
- **Endpoint:** `/api/ai/context`
- **Purpose:** Single-call executive telemetry snapshot returning active subsidiaries, pending fund requests, budget utilization, attendance summary, payroll status, intercompany balances, and pending staged actions.

4. Complete Reference: All 16 BEMI AI Tool Endpoints

Tool 1: Fund Request Approvals (`/api/ai/tools/approvals`)

Purpose: Inspect the pending fund request queue and stage executive decisions (`APPROVE`, `REJECT`, or `HOLD`) with automatic beneficiary bank and USSD quick-pay metadata.

A. List Pending Fund Requests (`GET`)

- **URL:** `GET /api/ai/tools/approvals?status=PENDING&limit=20`
- **Optional Query Params:** `status` (`PENDING` | `APPROVED` | `REJECTED` | `ALL`), `subsidiaryId` , `limit`

B. Stage an Approval Decision (`POST`)

- **URL:** `POST /api/ai/tools/approvals`
- **Action Type Staged:** `approval`
- **Payload Schema:**

```JSON
{
  "requestId": "REQ-2026-0005",
  "decision": "APPROVE",
  "comment": "Budget verified; approved for immediate procurement."
}
```

- **Field Notes:**
- `requestId` *(required)*: Request number (`REQ-2026-XXXX`) or UUID.
- `decision` *(required)*: `"APPROVE"` , `"REJECT"` , or `"HOLD"` .
- `comment` *(optional)*: Executive review note saved on confirmation.
- **On Human Confirmation:** Transitions `FundRequest` status (`APPROVED` / `REJECTED` / `ON_HOLD`), records `ApprovalDecision` , updates budget commitment, and triggers requester notification.

Tool 2: Fund Requests, Recurring Templates & Clarifications (`/api/ai/tools/requests`)

Purpose: Query fund requests or recurring templates, and stage creating new fund requests, creating recurring expense schedules, posting thread messages, requesting formal clarification from staff, or deleting pending requests.

A. List Requests & Recurring Templates (`GET`)

- **URL:** `GET /api/ai/tools/requests?status=SUBMITTED&includeRecurring=true&limit=25`

B. Stage Creating a New Fund Request (`POST` — `CREATE_REQUEST`)

- **URL:** `POST /api/ai/tools/requests`
- **Action Type Staged:** `request_creation`
- **Payload:**

```JSON
{
  "action": "CREATE_REQUEST",
  "title": "Q4 Diesel Supply for HQ Generators",
  "description": "2,000 liters automotive gas oil for backup power",
  "category": "FACILITIES_MAINTENANCE",
  "amount": 2400000,
  "currency": "NGN",
  "urgency": "HIGH",
  "subsidiaryId": "BEMS-GROUP",
  "requesterId": "babatundediekolayomi@gmail.com",
  "vendorName": "Mikano International Ltd",
  "beneficiaryName": "Mikano International Ltd",
  "beneficiaryBank": "Guaranty Trust Bank",
  "beneficiaryAccount": "0123456789"
}
```

C. Stage Creating a Recurring Fund Request Template (`POST` — `CREATE_RECURRING`)

```JSON
{
  "action": "CREATE_RECURRING",
  "title": "Monthly Starlink Enterprise Subscription",
  "description": "High-speed satellite internet for HQ",
  "category": "IT_SOFTWARE_CLOUD",
  "amount": 150000,
  "frequency": "MONTHLY",
  "dayOfMonth": 1,
  "subsidiaryId": "BEMS-GROUP"
}
```

D. Stage Posting a Comment on a Request Thread (`POST` — `SEND_MESSAGE`)

```JSON
{
  "action": "SEND_MESSAGE",
  "requestId": "REQ-2026-0005",
  "message": "Please confirm if delivery is included in the vendor quote."
}
```

E. Stage Requesting Formal Clarification (`POST` — `REQUEST_CLARIFICATION`)

```JSON
{
  "action": "REQUEST_CLARIFICATION",
  "requestId": "REQ-2026-0005",
  "message": "Attach the proforma invoice from the vendor before final GMD sign-off."
}
```

- **On Human Confirmation:** Places the request in `NEEDS_CLARIFICATION` (`ON_HOLD`), posts the clarification message, and notifies the requester.

F. Stage Deleting a Pending Request (`POST` — `DELETE_REQUEST`)

```JSON
{
  "action": "DELETE_REQUEST",
  "requestId": "REQ-2026-0009",
  "reason": "Duplicate submission by requester"
}
```

Tool 3: Accountability & Retirement Governance (`/api/ai/tools/accountability`)

Purpose: Inspect open post-disbursement accountability cases (proof of spend / receipt retirement) and stage clearing a case, flagging a financial discrepancy, or extending a retirement deadline.

A. List Accountability Cases (`GET`)

- **URL:** `GET /api/ai/tools/accountability?status=OPEN`
- **Optional Query Params:** `status` (`OPEN` | `SUBMITTED` | `OVERDUE` | `FLAGGED` | `CLEARED`), `requestId`

B. Stage Clearing an Accountability Case (`POST` — `CLEAR`)

- **URL:** `POST /api/ai/tools/accountability`
- **Action Type Staged:** `accountability`
- **Payload:**

```JSON
{
  "decision": "CLEAR",
  "requestId": "REQ-2026-0005",
  "note": "Verified physical delivery waybill and vendor receipt."
}
```

C. Stage Flagging a Discrepancy (`POST` — `FLAG_DISCREPANCY`)

```JSON
{
  "decision": "FLAG_DISCREPANCY",
  "requestId": "REQ-2026-0005",
  "note": "Receipt total is ₦135,000 against ₦150,000 disbursed; unretired balance of ₦15,000 required."
}
```

D. Stage Extending an Accountability Deadline (`POST` — `EXTEND_DEADLINE`)

```JSON
{
  "decision": "EXTEND_DEADLINE",
  "requestId": "REQ-2026-0005",
  "extensionDays": 7,
  "note": "Vendor invoice delayed due to interstate transit."
}
```

Tool 4: Subsidiary & Category Budgets (`/api/ai/tools/budgets`)

Purpose: Query monthly subsidiary budgets and category utilization, or stage creating/updating a subsidiary's monthly budget and category allocation lines.

A. Query Subsidiary Budgets (`GET`)

- **URL:** `GET /api/ai/tools/budgets?subsidiary=BEMS-GROUP&month=2026-11`

B. Stage Monthly Budget Creation / Allocation Update (`POST`)

- **URL:** `POST /api/ai/tools/budgets`
- **Action Type Staged:** `budget`
- **Payload:**

```JSON
{
  "subsidiary": "BEMS-GROUP",
  "month": "2026-11",
  "totalBudget": 50000000,
  "allocations": [
    { "category": "OPERATIONS_LOGISTICS", "allocated": 20000000 },
    { "category": "IT_SOFTWARE_CLOUD", "allocated": 10000000 },
    { "category": "FACILITIES_MAINTENANCE", "allocated": 10000000 },
    { "category": "MARKETING_BRAND", "allocated": 5000000 },
    { "category": "STAFF_WELFARE_TRAINING", "allocated": 5000000 }
  ]
}
```

- **On Human Confirmation:** Upserts the `Budget` record and all `BudgetAllocation` lines for the specified subsidiary and period.

Tool 5: Payroll Lifecycle & Salary Adjustments (`/api/ai/tools/payroll`)

Purpose: Inspect monthly payroll batches and individual staff payslips, and stage base salary revisions, monthly payroll preparation, GMD batch sign-off, salary payout recording, removing an item from a draft batch, or confirming staff salary receipt.

A. List Payroll Periods or Payslip Items (`GET`)

- **Summary of all periods:** `GET /api/ai/tools/payroll`

- **Detailed payslips for a month:** `GET /api/ai/tools/payroll?action=GET_ITEMS&year=2026&month=11`

B. Stage Base Salary Adjustment (`POST` — `UPDATE_SALARY`)

- **URL:** `POST /api/ai/tools/payroll`
- **Action Type Staged:** `salary_change`
- **Payload:**

```JSON
{
  "action": "UPDATE_SALARY",
  "employeeId": "BEMS-BSA-2026-0002",
  "newSalary": 250000,
  "reason": "Annual performance review increment"
}
```

C. Stage Monthly Payroll Batch Preparation (`POST` — `PREPARE`)

- **Action Type Staged:** `payroll_action`

```JSON
{
  "action": "PREPARE",
  "year": 2026,
  "month": 11,
  "subsidiaryId": "BEMS-GROUP"
}
```

- **On Human Confirmation:** Calculates gross pay, aggregates attendance lateness/absence deductions, and creates/updates the `PayrollPeriod`, `PayrollBatch`, and `PayrollItem` rows.

D. Stage GMD Payroll Sign-Off (`POST` — `SIGN_OFF`)

```JSON
{
  "action": "SIGN_OFF",
  "year": 2026,
  "month": 11
}
```

E. Stage Salary Payout / Disbursement (`POST` — `PAYOUT` / `RECORD_PAYOUT`)

```JSON
{
  "action": "PAYOUT",
  "year": 2026,
  "month": 11,
  "itemId": "optional_single_payroll_item_uuid",
  "bankReference": "NIP-202611-00981"
}
```

F. Stage Removing a Staff Payslip from a Draft Batch (`POST` — `REMOVE_ITEM`)

```JSON
{
  "action": "REMOVE_ITEM",
  "itemId": "payroll_item_uuid",
  "reason": "Employee on unpaid sabbatical"
}
```

G. Stage Confirming Staff Salary Receipt (`POST` — `CONFIRM_RECEIPT`)

```JSON
{
  "action": "CONFIRM_RECEIPT",
  "itemId": "payroll_item_uuid",
  "note": "Confirmed credit alert received by staff"
}
```

Tool 6: Staff Directory & Profile Governance (`/api/ai/tools/staff`)

Purpose: Search the staff directory and stage any staff administration operation: editing profile fields (name, email, phone, job title, department, subsidiary transfer, role), toggling attendance exemptions, suspending/freezing/activating a staff profile, registering an offline staff profile, requesting a profile update, or deactivating/deleting a staff record.

A. List / Search Staff Directory (`GET`)

- **URL:** `GET /api/ai/tools/staff?search=Babatunde&subsidiaryId=BEMS-GROUP&status=ACTIVE`

B. Stage Editing Staff Profile Data (`POST` — `EDIT_PROFILE`)

- **URL:** `POST /api/ai/tools/staff`
- **Action Type Staged:** `staff_management`
- **Payload:**

```JSON
{
  "action": "EDIT_PROFILE",
  "employeeId": "BEMS-BSA-2026-0002",
  "legalName": "Babatunde Diekolayomi",
  "workEmail": "babatunde@bemsgroup.com",
  "phone": "+2348031234567",
  "jobTitle": "Senior Operations Lead",
  "department": "Operations",
  "subsidiaryId": "BEMS-GROUP",
  "role": "STAFF"
}
```

- **Note on `role` :** Must strictly be `"STAFF"` , `"GMD"` , or `"SUPER_ADMIN"` .

C. Stage Toggling Attendance Exemption (`POST` — `TOGGLE_ATTENDANCE_EXEMPT`)

```JSON
{
  "action": "TOGGLE_ATTENDANCE_EXEMPT",
  "employeeId": "BEMS-BSA-2026-0002",
  "attendanceExempt": true,
  "reason": "Field engineering role with remote deployment"
}
```

D. Stage Changing Employment / Account Status (`POST` — `CHANGE_STATUS`)

```JSON
{
  "action": "CHANGE_STATUS",
  "employeeId": "BEMS-BSA-2026-0002",
  "status": "SUSPENDED",
  "reason": "Pending internal audit review"
}
```

- **Allowed `status` values:** `"ACTIVE"`, `"SUSPENDED"`, `"FROZEN"`, `"INACTIVE"`, `"TERMINATED"`.

E. Stage Offline Staff Registration (`POST` — `REGISTER_OFFLINE`)

```JSON
{
  "action": "REGISTER_OFFLINE",
  "legalName": "Musa Abdullahi",
  "workEmail": "musa.abdullahi@bemsgroup.com",
  "phone": "+2348099887766",
  "jobTitle": "Site Technician",
  "department": "Engineering",
  "subsidiaryId": "BEMS-GROUP",
  "baseSalary": 120000
}
```

F. Stage Requesting a Staff Profile Update (`POST` — `REQUEST_UPDATE`)

```JSON
{
  "action": "REQUEST_UPDATE",
  "employeeId": "BEMS-BSA-2026-0002",
  "message": "Please update your emergency contact and tax ID on the Staff Portal."
}
```

G. Stage Deactivating or Deleting a Staff Profile (`POST` — `DELETE_OR_DEACTIVATE`)

```JSON
{
  "action": "DELETE_OR_DEACTIVATE",
  "employeeId": "BEMS-BSA-2026-0002",
  "reason": "Duplicate test account cleanup"
}
```

Tool 7: Executive Admin Onboarding (`/api/ai/tools/admin-onboarding`)

Purpose: Manage executive onboarding invitations strictly for `SUPER_ADMIN` and `GMD` (Staff onboard via self-service registration on the Staff Portal / Mobile App).

A. List Executive Invitations (`GET`)

- **URL:** `GET /api/ai/tools/admin-onboarding`

B. Stage New Executive Invitation (`POST` — `INVITE`)

- **URL:** `POST /api/ai/tools/admin-onboarding`
- **Action Type Staged:** `admin_onboarding`
- **Payload:**

```JSON
{
  "action": "INVITE",
  "legalName": "Engr. Michael Adeyemi",
  "workEmail": "adeyemi@bemsgroup.com",
  "phone": "+2348030001122",
  "role": "SUPER_ADMIN",
  "subsidiaryId": "BEMS-GROUP"
}
```

C. Stage Resending an Executive Invitation (`POST` — `RESEND_INVITE`)

```JSON
{
  "action": "RESEND_INVITE",
  "invitationId": "inv_uuid_or_work_email"
}
```

D. Stage Revoking an Executive Invitation (`POST` — `REVOKE_INVITE`)

```JSON
{
  "action": "REVOKE_INVITE",
  "invitationId": "inv_uuid_or_work_email",
  "reason": "Incorrect email address entered"
}
```

Tool 8: Corporate Subsidiary Management (`/api/ai/tools/subsidiary`)

Purpose: List active subsidiaries across the conglomerate, stage creating a new subsidiary, or stage editing an existing subsidiary's metadata, RC number, headquarters, industry, brand color, active status, and monthly budget.

A. List Subsidiaries (`GET`)

- **URL:** `GET /api/ai/tools/subsidiary`

B. Stage Creating a New Subsidiary (`POST` — `CREATE`)

- **URL:** `POST /api/ai/tools/subsidiary`
- **Action Type Staged:** `subsidiary_creation`
- **Payload:**

```JSON
{
  "action": "CREATE",
  "name": "Studio Arella Ltd",
  "code": "SAL",
  "type": "SUBSIDIARY",
  "currency": "NGN",
  "brandColor": "#EA580C",
  "industry": "Media & Creative Production",
  "headquarters": "Lekki Phase 1, Lagos"
}
```

C. Stage Editing an Existing Subsidiary (`POST` — `UPDATE`)

- **Action Type Staged:** `subsidiary_update`
- **Payload:**

```JSON
{
  "action": "UPDATE",
  "subsidiaryId": "SAL",
  "name": "Studio Arella Media Ltd",
  "code": "SAML",
  "brandColor": "#2563EB",
  "industry": "Digital Media & Broadcast",
  "headquarters": "Victoria Island, Lagos",
  "rcNumber": "RC-1928374",
  "monthlyBudget": 35000000,
  "isActive": true
}
```

Tool 9: Attendance, Shift Rosters & Geofence Stations (`/api/ai/tools/attendance`)

Purpose: Inspect daily attendance logs and shift policies, and stage excusing lateness/absence, overriding attendance records, manual clock-in/out, updating global shift & Starlink/GPS geofence enforcement policies, updating individual staff work schedule rosters, or calibrating attendance station GPS coordinates.

A. Get Live Attendance Log & Policies (`GET`)

- **URL:** `GET /api/ai/tools/attendance?date=2026-11-06`

B. Stage Excusing Lateness or Absence (`POST` — `EXCUSE`)

- **URL:** `POST /api/ai/tools/attendance`
- **Action Type Staged:** `attendance_override`

```JSON
{
  "action": "EXCUSE",
  "employeeId": "BEMS-BSA-2026-0002",
  "workDate": "2026-11-06",
  "reason": "Official client meeting at Victoria Island"
}
```

C. Stage Overriding an Attendance Record (`POST` — `OVERRIDE`)

```JSON
{
  "action": "OVERRIDE",
  "employeeId": "BEMS-BSA-2026-0002",
  "workDate": "2026-11-06",
  "status": "PRESENT",
  "isLate": false,
  "deductionPercent": 0,
  "reason": "Biometric kiosk sync delay"
}
```

D. Stage Manual Clock-In or Clock-Out (`POST` — `MANUAL_CLOCK_IN` / `MANUAL_CLOCK_OUT`)

```JSON
{
  "action": "MANUAL_CLOCK_IN",
  "employeeId": "BEMS-BSA-2026-0002",
  "workDate": "2026-11-06",
  "clockInTime": "08:05",
  "reason": "Verified arrival at HQ reception"
}
```

E. Stage Updating Shift & Starlink/GPS Geofence Policy (`POST` — `UPDATE_POLICY`)

```JSON
{
  "action": "UPDATE_POLICY",
  "resumptionTime": "08:00",
  "closingTime": "17:00",
  "gracePeriodMinutes": 15,
  "latenessDeductionPercent": 5.0,
  "requireStarlinkWifi": true,
  "requireGpsGeofence": true,
  "allowedSsid": "BEMS_STARLINK_HQ"
}
```

F. Stage Updating a Staff Member's Work Schedule Roster (`POST` — `UPDATE_ROSTER`)

```JSON
{
  "action": "UPDATE_ROSTER",
  "employeeId": "BEMS-BSA-2026-0002",
  "expectedClockIn": "08:30",
  "expectedClockOut": "17:30",
  "gracePeriodMinutes": 20,
  "offDays": "0,6",
  "notes": "Adjusted shift for Lagos mainland commute"
}
```

G. Stage Calibrating an Attendance Station's GPS Geofence (`POST` — `CALIBRATE_STATION`)

```JSON
{
  "action": "CALIBRATE_STATION",
  "name": "BEMS HQ Main Gate",
  "latitude": 6.5244,
  "longitude": 3.3792,
  "radiusMeters": 150,
  "allowedWifiSsid": "BEMS_STARLINK_HQ",
  "starlinkIpRanges": "192.168.1.0/24"
}
```

Tool 10: HR Leave Management (`/api/ai/tools/hr-leave`)

Purpose: Query staff leave applications and stage filing a new leave request or approving/rejecting an existing leave application.

A. List Leave Applications (`GET`)

- **URL:** `GET /api/ai/tools/hr-leave?status=PENDING`

B. Stage Creating a Leave Application (`POST` — `CREATE`)

- **URL:** `POST /api/ai/tools/hr-leave`
- **Action Type Staged:** `leave_action`

```JSON
{
  "action": "CREATE",
  "employeeId": "BEMS-BSA-2026-0002",
  "leaveType": "ANNUAL",
  "startDate": "2026-12-15",
  "endDate": "2026-12-22",
  "reason": "Annual family vacation"
}
```

C. Stage Approving or Rejecting a Leave Request (`POST` — `APPROVE` / `REJECT`)

```JSON
{
  "action": "APPROVE",
  "leaveId": "leave_request_uuid",
  "reviewNote": "Approved by GMD; handover coverage confirmed."
}
```

Tool 11: Vendors, GRNs, Invoices & 3-Way Match (`/api/ai/tools/vendors`)

Purpose: Manage the verified supplier registry and complete 3-way matching pipeline (Purchase Order / Fund Request + Goods Receipt Note + Vendor Invoice).

A. List Vendors or Inspect Read-Only 3-Way Match Status (`GET`)

- **List Vendors:** `GET /api/ai/tools/vendors`
- **Read-Only 3-Way Match Inspection:** `GET /api/ai/tools/vendors?action=CHECK_MATCH&requestId=REQ-2026-0005`

B. Stage Registering a Verified Vendor (`POST` — `REGISTER`)

- **URL:** `POST /api/ai/tools/vendors`
- **Action Type Staged:** `vendor_registration`

```JSON
{
  "action": "REGISTER",
  "name": "Mikano International Ltd",
  "category": "FACILITIES",
  "bankName": "Guaranty Trust Bank",
  "accountNumber": "0123456789",
  "accountName": "Mikano International Ltd",
  "taxId": "TIN-99283746",
  "contactPhone": "+2348020003344",
  "contactEmail": "billing@mikano.com"
}
```

C. Stage Recording a Goods Receipt Note (`POST` — `CREATE_GRN`)

- **Action Type Staged:** `grn_creation`

```JSON
{
  "action": "CREATE_GRN",
  "requestId": "REQ-2026-0005",
  "waybillNumber": "WB-2026-8841",
  "status": "ACCEPTED",
  "inspectionNotes": "All 4 Starlink kits delivered sealed and tested."
}
```

- **Allowed `status` values:** `"ACCEPTED"`, `"PARTIAL"`, `"DAMAGED"`.

D. Stage Logging a Vendor Invoice (`POST` — `CREATE_INVOICE`)

- **Action Type Staged:** `vendor_invoice_creation`

```JSON
{
  "action": "CREATE_INVOICE",
  "requestId": "REQ-2026-0005",
  "vendorId": "Mikano International Ltd",
  "invoiceNumber": "INV-MIK-2026-901",
  "amount": 150000,
  "tax": 11250,
  "currency": "NGN",
  "invoiceDate": "2026-11-06"
}
```

E. Stage 3-Way Match Verification & State Update (`POST` — `STAGE_MATCH`)

- **Action Type Staged:** `three_way_match`

```JSON
{
  "action": "STAGE_MATCH",
  "requestId": "REQ-2026-0005"
}
```

- **On Human Confirmation:** Runs the 3-way match engine across the PO (`FundRequest`), `GoodsReceiptNote` , and `VendorInvoice` , and persists the resulting `MATCHED` / `MISMATCH_FLAGGED` state.

Tool 12: Intercompany Debit/Credit Ledger & Netting (`/api/ai/tools/intercompany`)

Purpose: Inspect bilateral intercompany balances across subsidiaries and stage paired Debit/Credit note generation or netting settlement.

A. Query Conglomerate Netting Matrix (`GET`)

- **URL:** `GET /api/ai/tools/intercompany`

B. Stage Paired Intercompany Debit/Credit Notes (`POST` — `CREATE_NOTE`)

- **URL:** `POST /api/ai/tools/intercompany`
- **Action Type Staged:** `intercompany_Entry`

```JSON
{
  "action": "CREATE_NOTE",
  "source": "BEMS-GROUP",
  "target": "SAL",
  "amount": 350000,
  "description": "Cross-charge: Shared Starlink & Cloud Infrastructure Q4"
}
```

C. Stage Intercompany Netting Settlement (`POST` — `STAGE_SETTLEMENT`)

- **Action Type Staged:** `intercompany_settle`

```JSON
{
  "action": "STAGE_SETTLEMENT",
  "noteIds": ["ic_note_uuid_1", "ic_note_uuid_2"],
  "settlementReference": "IC-NET-2026-11"
}
```

Tool 13: Bank Account Registry (`/api/ai/tools/bank-accounts`)

Purpose: Stage registering or updating verified 10-digit NUBAN bank details for **Staff** (`staff_bank_update`), **Subsidiaries / Treasury** (`treasury_bank_register`), or **Vendors** (`vendor_bank_update`).

A. List Registered Bank Accounts (`GET`)

- **URL:** `GET /api/ai/tools/bank-accounts?targetType=STAFF`

B. Stage Staff, Subsidiary, or Vendor Bank Update (`POST`)

- **URL:** `POST /api/ai/tools/bank-accounts`
- **Payload (Staff Example):**

```JSON
{
  "targetType": "STAFF",
  "targetId": "BEMS-BSA-2026-0002",
  "bankName": "Kuda Microfinance Bank",
  "accountNumber": "2001928374",
  "accountName": "Babatunde Diekolayomi"
}
```

- **Payload (Subsidiary Treasury Example):**

```JSON
{
  "targetType": "SUBSIDIARY",
  "targetId": "BEMS-GROUP",
  "bankName": "Zenith Bank",
  "accountNumber": "1019283746",
  "accountName": "BEMS Group Holdings Operational",
  "accountType": "Corporate Operational"
}
```

Tool 14: Executive Follow-Up Tasks (`/api/ai/tools/tasks`)

Purpose: Track and stage executive follow-up tasks assigned to staff or admins.

A. List Follow-Up Tasks (`GET`)

- **URL:** `GET /api/ai/tools/tasks?status=PENDING`

B. Stage Creating a Follow-Up Task (`POST` — `CREATE`)

- **URL:** `POST /api/ai/tools/tasks`
- **Action Type Staged:** `task_action`

```JSON
{
  "action": "CREATE",
  "title": "Submit Q4 Vendor Compliance Certificates",
  "description": "Collect updated CAC and TIN documents for top 5 logistics vendors.",
  "assignedToId": "BEMS-BSA-2026-0002",
  "priority": "HIGH",
  "dueDate": "2026-11-15"
}
```

C. Stage Completing or Updating a Task (`POST` — `COMPLETE` / `UPDATE_STATUS`)

```JSON
{
  "action": "COMPLETE",
  "taskId": "task_uuid",
  "note": "All certificates uploaded and verified."
}
```

Tool 15: Direct Staff Helpdesk Messaging (`/api/ai/tools/messages`)

Purpose: Read Staff-Admin helpdesk threads and stage direct GMD or HR messages to any staff member (creates a `StaffAdminMessage`, in-app `SystemNotification`, and email alert on confirmation).

A. List Helpdesk Messages (`GET`)

- **URL:** `GET /api/ai/tools/messages?channel=GMD&employeeId=optional_uuid`

B. Stage Direct Executive Message to Staff (`POST`)

- **URL:** `POST /api/ai/tools/messages`
- **Action Type Staged:** `staff_message`

```JSON
{
  "employeeId": "BEMS-BSA-2026-0002",
  "channel": "GMD",
  "message": "Please upload the signed delivery waybill for REQ-2026-0005 before 4 PM today."
}
```

Tool 16: Approval Policies & Workspace Governance Settings (`/api/ai/tools/policies`)

Purpose: Inspect or stage changes to conglomerate approval policy bands (`ApprovalPolicy`) and workspace-level governance settings (`WorkspaceSetting`).

A. List Approval Policies & Workspace Settings (`GET`)

- **URL:** `GET /api/ai/tools/policies?organizationId=BEMS-GROUP`

B. Stage Creating or Updating an Approval Policy Band (`POST` — `UPSERT_POLICY`)

- **URL:** `POST /api/ai/tools/policies`
- **Action Type Staged:** `policy_update`

```JSON
{
  "action": "UPSERT_POLICY",
  "organizationId": "BEMS-GROUP",
  "requestCategory": "CAPITAL_EXPENDITURE",
  "minAmountMinor": 0,
  "maxAmountMinor": 5000000000,
  "requiredRoles": ["GMD"],
  "slaHours": 24,
  "escalationRole": "GMD",
  "autoApproveBelowMinor": 0
}
```

C. Stage Updating Workspace Governance Settings (`POST` — `UPDATE_WORKSPACE_SETTINGS`)

```JSON
{
  "action": "UPDATE_WORKSPACE_SETTINGS",
  "organizationId": "BEMS-GROUP",
  "settingKey": "approval.controls",
  "settings": {
    "SLA_REMINDER_HOURS": 12,
    "REQUIRE_THREE_WAY_MATCH_ABOVE_NGN": 500000
  }
}
```

---

5. Human Confirmation & Cancellation Protocol

Once n8n or BEMI stages an action via any `POST /api/ai/tools/*` route, the proposal appears in:

1. The **BEMI Copilot Staged Action Card** in the web dashboard / mobile app.
2. `GET /api/ai/staged-actions` (pending proposals queue).

5.1 Confirm & Execute Staged Action (Human Browser Session Only)

- **Method:** `POST`
- **Endpoint:** `/api/ai/staged-actions/{actionId}/confirm`

- **Authentication:** Authenticated Human Session Cookie (`bems_session`) — **Service tokens (`x-api-key` / `BEMS_AI_SERVICE_TOKEN`) are strictly blocked with `403 Forbidden`.**
- **Optional Request Body:**

```
JSON
{
  "payloadHash": "sha256:8f9d2a..."
}
```

- **Execution Guarantees:**
- Atomically locks the `StagedAction` row (`PENDING` -> `EXECUTING` -> `EXECUTED`).
- Verifies the `payloadHash` and checks that `expiresAt` has not passed.
- Verifies the human user's role (`GMD` / `SUPER_ADMIN` / `STAFF`) against the specific action's permission gate.
- Executes the canonical database transaction and records an immutable `AuditEvent`.

5.2 Cancel / Reject Staged Action (Human Browser Session Only)

- **Method:** `POST`
- **Endpoint:** `/api/ai/staged-actions/{actionId}/cancel`
- **Authentication:** Authenticated Human Session Cookie (`bems_session`).

---

6. Ready-to-Import n8n HTTP Request Node Template

Paste this JSON directly into your n8n workflow canvas to create a pre-configured **BEMS AI Tool Stage Request** node:

JSON


```
{
  "parameters": {
    "method": "POST",
    "url": "https://bemserp.work/api/ai/tools/approvals",
    "sendHeaders": true,
    "headerParameters": {
      "parameters": [
        {
          "name": "Authorization",
          "value": "Bearer {{ $env.BEMS_AI_SERVICE_TOKEN }}"
        },
        {
          "name": "x-actor-email",
          "value": "={{ $json.executiveEmail || 'gmd@bemsgroup.com' }}"
        },
        {
          "name": "Content-Type",
          "value": "application/json"
        }
      ]
    },
    "sendBody": true,
    "specifyBody": "json",
    "jsonBody": "={\n  \"requestId\": \"{{ $json.requestNumber }}\",\n  \"decision\": \"{{
$json.decision || 'APPROVE' }}\",\n  \"comment\": \"{{ $json.comment || 'Staged via n8n AI
Automation' }}\"\n}",
    "options": {}
  },
  "name": "BEMS Stage AI Action",
  "type": "n8n-nodes-base.httpRequest",
  "typeVersion": 4.2
}
```

7. Error Codes & Troubleshooting for Automation Engineers

HTTP Status	Error Code / Condition	Cause & Resolution
`400 Bad Request`	`INCOMPLETE_PARAMETERS`	Required fields were omitted. Inspect `missing_fields` and `conversational_prompt` in the response JSON and prompt the user for the missing values (never invent dummy data).
`401 Unauthorized`	Missing / Invalid Token	Ensure `Authorization: Bearer <BEMS_AI_SERVICE_TOKEN>` or `x-api-key: <BEMS_AI_SERVICE_TOKEN>` matches the server `.env`.
`403 Forbidden`	Service Token on `/confirm`	Automation service tokens cannot confirm staged actions. Present the staged action card or dashboard link so the human executive can click **Confirm**.
`404 Not Found`	Entity Not Found	The target `requestId`, `employeeId`, `subsidiaryId`, or `vendorId` could not be resolved. Call the corresponding `GET /api/ai/tools/*` endpoint first to look up the exact ID or code.
`409 Conflict`	Already Executed / Expired	The staged action was already confirmed, cancelled, or passed its 30-minute `expiresAt` window. Stage a fresh proposal if still needed.